

ENGINEERING BULLETIN

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Subject : Features for K Smart Controller for Screw Chillers

Features of 'K Smart' Controller for Kirloskar Screw Chillers

A) The Main controller for the chiller consists of

- a. CPU H8S2320, 16 bit, 24MHz.
- b. Program Memory 2+2 MB , 16 bit
- c. Data memory(RAM) 512 KB, 16bit
- d. Permanent memory 13 KB
- e. Working cycle duration approx. 200 ms

B) The following data shall be available on display of the controller

1. Suction Pressure
2. Discharge Pressure
3. Chilled Water Inlet Temp
4. Chilled Water Outlet Temp
5. Condenser Water Inlet Temp
6. Condenser Water Outlet Temp
7. Liquid Line Refrigerant Temp
8. Suction Line Refrigerant Temp.
9. Saturated Suction Temp.
10. Suction Super Heat
11. Saturated Discharge Temp.
12. Discharge line Refrigerant Temp.
13. Discharge Super Heat
14. Compressor Motor Current in Amp (only one phase)
15. Compressor Motor Current in % RLA
16. Date & Time
17. Compressor On / Off
18. Water flow ON/OFF indication (Condenser)
19. Water flow ON/OFF indication (Evaporator)
20. Oil temperature (for KW4, KWF & KA4 chiller models only)

Kirloskar Chillers Private Limited

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21. Oil feed pressure (for KW4, KWF & KA4 chiller models only)
22. Differential Oil pressure
23. Motor winding temperature – Healthy or Un-healthy
24. Voltage Protection Device – Healthy or Un-healthy
25. EXV opening %

C) Safeties: Following safeties are directly sensed by the controller & acted upon

1. High Condensing Pressure
2. Low Condensing pressure (for KAS, & KA4 chiller models only)
3. Low Suction Pressure
4. Low Oil Pressure Differential
5. Antifreeze Trip.
6. Compressor Motor Overload on Current. (High Amps)
7. Compressor Motor Under load on Current (Low Amps.)
8. High Oil Temperature (for KW4, KWF & KA4 chiller models only)
9. Low Oil Level (for KWF chiller models only)
10. Pro Active unload for

- High Current
- High Discharge Pressure
- Low Suction Pressure
- High Discharge Temperature
- High Oil Temperature

D) Other safeties which are sensed by other devices, but action is taken by the controller:

- Over Voltage / Under Voltage / Single phasing / Reverse phasing / Voltage unbalance.
- Motor Winding Overheat.
- Low Condenser Water Flow (flow switch is under clients scope)
- Low Chilled Water Flow (flow switch is under clients scope)

E) Other Features:

1. The controller display consists of a LCD Display with backlight, with IP65 protection and 6 silicon keys with LEDs. The display is a semi graphic type of display with resolution of 120x32 pixels. The display is a 4 rows x 20 columns display with the font size of 5 x 7
2. The controller is equipped with battery backup for memory and clock to take care of any power failures
3. The controller is programmed for automatic start up on resumption of power after power failure.

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4. The controller has a in built compressor lead-lag and sequencing facility for multi compressor units, which ensures proper starting and stopping of compressors as well as equal start stops and run hours for all the compressors.
5. The controller has an internal 7-day, 8 holiday clock with programmable time scheduling with Interval / RTC (In time scheduling operator can specify week days as per mention. each weekday schedule will be followed on that particular weekday and the holiday schedule is followed on the days marked as holiday. For example one can have schedule as below:

Monday: Sch 1 - 7:00 to 12:30 – set point 1
Sch 2 - 13:30 to 18:00 – set point 2

Tuesday: Sch 1 - 7:00 to 12:30– set point 1
Sch 2 - 13:30 to 18:00– set point 2

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Saturday: Sch1 – 7:00 to 12:30– set point 1
Sch2 – 0:00 to 0:00– set point 2

Sunday: Sch1 – 0:00 to 0:00
Sch2 – 0:00 to 0:00

Holiday: Sch1 – 0:00 to 0:00
Sch2 – 0:00 to 0:00

Holidays: 3rd Jan
14th Feb
21st March
30th Oct

Now it will follow schedule of each weekdays but on specified holidays it will follow schedule mentioned for holiday.)

6. The controller has a detailed alarm history :- The controller logs last 99 alarms with 4 Variables for each alarm (critical parameters) and these can be viewed on the controller display
7. The anti recycle timer (Start to Start / Stop to Start) is based on the controllers inbuilt RTC
8. Important variables list incorporated for getting connected to BMS system
9. Access to all set points with 3 Levels of Password – User, Supervisory and Factory
10. Alarm and Data Logging can be downloaded in CSV (comma separated variables) format
11. Multi Chiller sequencing: - This is useful where user has more than one chiller at site and would like to have one (or more) as standby as well as run time equalization among all chillers. For example a site has 3 chillers and customer wants 2 working + 1 standby. With 'K Smart' controller in all chillers we need no external hardware apart from wiring to handle the situation. With unique addresses given to all controller and setting some parameters (like rotation time = 6hours) at any given time only 2 chillers will run. Standby will get rotated after every 6 hours (so that each chiller will run for 12 hours and will be at standby for 6 hours). In case of serious alarm in any working chiller standby will take its place immediately.

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